SCREENSHOT OF SPARK SESSION

```python
# Spark Session
from pyspark.sql import SparkSession

spark = SparkSession.builder \
.appName("Distributed Data Processing Activity") \
.getOrCreate()

spark
```

SparkSession - in-memory

SparkContext

Spark UI

Version
v4.0.2
Master
local[*]
AppName
Distributed Data Processing Activity

OUTPUT SCREENSHOTS

```python
# Load dataset

from pyspark.sql.functions import col

df = spark.read.csv("phivolcs_earthquake_data.csv", header=True, inferSchema=True)
df.show(5)
```

```
+-------------------+--------+---------+-----------+---------+--------------------+-----------------+----------------+
|       Date_Time_PH|Latitude|Longitude|Depth_In_Km|Magnitude|            Location|Specific_Location|General_Location|
+-------------------+--------+---------+-----------+---------+--------------------+-----------------+----------------+
|2026-02-05 12:43:00|    6.43|   123.69|        3.0|      2.5|042 km S 70° W of...|      Kalamansig |  Sultan Kudarat|
|2026-02-05 12:36:00|     6.4|   123.73|        8.0|      3.5|040 km S 64° W of...|      Kalamansig |  Sultan Kudarat|
|2026-02-05 12:22:00|     6.3|   123.68|       21.0|      2.4|050 km S 56° W of...|      Kalamansig |  Sultan Kudarat|
|2026-02-05 12:01:00|    6.67|   124.29|       13.0|      1.6|025 km N 82° E of...|           Lebak |  Sultan Kudarat|
|2026-02-05 11:18:00|   14.04|   122.27|        3.0|      3.6|009 km N 10° W of...|         Calauag |          Quezon|
+-------------------+--------+---------+-----------+---------+--------------------+-----------------+----------------+
only showing top 5 rows
```

- Is the data processed sequentially or distributed?

  The data is processed in a distributed way. From what I know, Spark divides the data in smaller portions and processes them across multiple cores.

- What does Spark do internally?

  Spark breaks data into smaller portions, and distributes it to multiple processing units or cores. Afterwards, the operations it starts to do is parallel.

Filter Data

```
filtered_df = df.filter(col("Magnitude") > 3)
filtered_df.show()
```

```
+-------------------+--------+---------+-----------+---------+--------------------+-----------------+----------------+
|       Date_Time_PH|Latitude|Longitude|Depth_In_Km|Magnitude|            Location|Specific_Location|General_Location|
+-------------------+--------+---------+-----------+---------+--------------------+-----------------+----------------+
|2026-02-05 12:36:00|     6.4|   123.73|        8.0|      3.5|040 km S 64° W of...|      Kalamansig |  Sultan Kudarat|
|2026-02-05 11:18:00|   14.04|   122.27|        3.0|      3.6|009 km N 10° W of...|         Calauag |          Quezon|
|2026-02-05 05:34:00|    6.27|   123.79|       21.0|      4.6|043 km S 42° W of...|      Kalamansig |  Sultan Kudarat|
|2026-02-05 03:13:00|    7.34|   127.31|       17.0|      3.4|086 km N 81° E of...|           Manay |  Davao Oriental|
|2026-02-05 02:40:00|    4.77|   124.44|       33.0|      3.3|130 km S 57° W of...|    Balut Island |Davao Occidental|
|2026-02-05 02:34:00|    7.24|   127.08|        9.0|      3.6|060 km N 87° E of...|           Manay |  Davao Oriental|
|2026-02-05 02:03:00|    3.94|   126.57|       12.0|      3.4|207 km S 38° E of...|    Balut Island |Davao Occidental|
|2026-02-05 01:46:00|   15.72|   121.99|        9.0|      3.8|046 km S 84° E of...|           Baler |          Aurora|
|2026-02-05 01:33:00|   18.12|   119.94|       10.0|      4.2|060 km N 70° W of...|           Badoc |    Ilocos Norte|
|2026-02-05 01:30:00|   18.11|   119.93|       14.0|      3.1|059 km N 81° W of...|        Currimao |    Ilocos Norte|
|2026-02-05 01:27:00|   15.71|   122.01|       12.0|      3.5|048 km S 84° E of...|           Baler |          Aurora|
|2026-02-05 01:16:00|    7.26|   127.05|        9.0|      3.4|053 km S 82° E of...|          Caraga |  Davao Oriental|
|2026-02-04 21:14:00|    12.1|   123.92|        5.0|      4.5|014 km N 35° W of...|       Cataingan |         Masbate|
|2026-02-04 21:07:00|    6.42|   123.72|       16.0|      3.4|040 km S 68° W of...|      Kalamansig |  Sultan Kudarat|
|2026-02-04 20:27:00|    6.41|   123.69|        4.0|      3.2|043 km S 68° W of...|      Kalamansig |  Sultan Kudarat|
|2026-02-04 15:11:00|     9.3|   126.52|        7.0|      4.6|036 km N 86° E of...|          Cortes | Surigao Del Sur|
|2026-02-04 13:49:00|    6.46|    124.6|        9.0|      3.1|009 km N 66° W of...|      Santo Niño |  South Cotabato|
|2026-02-04 13:07:00|   15.12|   123.13|       22.0|      3.2|074 km N 15° E of...|   Tinaga Island | Camarines Norte|
|2026-02-04 03:34:00|    6.37|   123.76|        3.0|      3.1|038 km S 58° W of...|      Kalamansig |  Sultan Kudarat|
|2026-02-04 01:16:00|    4.44|   125.21|       20.0|      3.6|110 km S 12° W of...|    Balut Island |Davao Occidental|
+-------------------+--------+---------+-----------+---------+--------------------+-----------------+----------------+
only showing top 20 rows
```

Group and Aggregate

```
grouped_df = df.groupBy("Specific_Location").count()
grouped_df.show()
```

```
+-----------------+-----+
|Specific_Location|count|
+-----------------+-----+
|        Pambujan |  188|
|       Lambayong |   70|
|         Cabagan |    1|
|        Palompon |   30|
|        Bugallon |    7|
|          Panaon |    3|
|      Dinalungan |  258|
|        Pagbilao |    3|
|         Rosales |    4|
|        Lidlidda |    9|
|        Claveria |  566|
|     San Mariano |   10|
|           Bobon |   20|
|          Culaba |    5|
| City Of Calamba |    1|
|       Balilihan |   22|
|       Calatagan | 3611|
|     San Teodoro |   12|
|       Paracelis |   78|
|         Sirawai |   25|
+-----------------+-----+
only showing top 20 rows
```

Sort Results

```python
sorted_df = grouped_df.orderBy("count", ascending=False)
sorted_df.show()
```

```
+-------------------+-----+
|  Specific_Location|count|
+-------------------+-----+
|          Hinatuan | 4543|
|      City Of Bogo | 3996|
|      General Luna | 3619|
|         Calatagan | 3611|
| Governor Generoso | 3552|
|            Burgos | 3313|
|  Jose Abad Santos | 3015|
|            Lingig | 2608|
|             Manay | 2388|
|           Cagwait | 2225|
|         Sarangani | 1985|
|         Marihatag | 1888|
|            Cortes | 1781|
|            Lubang | 1614|
|         Tarragona | 1548|
|           Bayabas | 1530|
|           Tulunan | 1475|
|      Balut Island | 1442|
|        Kalamansig | 1407|
|           Baganga | 1111|
+-------------------+-----+
only showing top 20 rows
```

SCREENSHOTS OF ANALYSIS/QUESTION RESPONSES (all question answers are also included in the notebook)

PART 3 PERFORMANCE AND SCALABILITY DISCUSSION

1. How does Spark divide data?

   It divides data by getting smaller chunks of the dataset, called *partitions*.

2. What happens if dataset size increases?

   I believe that the processing remains the same, but may be slower. But, it can scale if it has more nodes, along with upgraded memory and resources.

3. Why is this better than standard Python processing?

   It is better since Python normally processes data sequentially and has memory limitations, while PySpark uses parallel. From what I know, parallel processing can be much more efficient in larger datasets.

PART 4 CLOUD COMPUTING ANALYSIS

1. Why is Google Colab considered cloud computing?

   It is because it runs on servers rather than locally, allowing users to access it directly on the web browser without needing to install applications or manage hardware.

2. Which cloud service model does this resemble (IaaS, PaaS, SaaS)?

   Google Colab resembles a PaaS (Platform as a Service) model since it is immediately accessible for coding and data processing. Additionally, it also contains pre-configured environments such as Python and a lot of libraries.

3. How does cloud infrastructure help distributed systems?

   Based from this activity, cloud servers can help by allowing easy scalability with adding more machines, and it has the ability to do parallel processing across multiple servers. Plus, its aforementioned high accessibility.